Exp No: 11

Develop a program to create a Reverse TCP Connection using Python sockets.

Aim:

To develop and execute a python program to establish a TCP connection between a listener and a connector using sockets and exchange messages over local host

Algorithm:

listener program:

1) Import the socket module
2) Define the host as 127.0.0.1 and port as 5000
3) Create a TCP socket
4) Bind the socket to the host and port
5) Listen for an incoming connection
6) Accept the connection and display the connector's address.
7) Receive and display the connector's message
8) Send a reply and close the connection and server socket

Connector program

1) Import the socket module
2) Define the host as 127.0.0.1 and port as 5000
3) Create a TCP socket and connect to the listener and send message to listener
4) Receive and display the listener's reply and close the client socket.

Procedure:

Code: listener program - listener.py

```
import socket
HOST = "127.0.0.1"
PORT = 5000
```

Output:

Listener o/p:

Waiting for reverse connection...

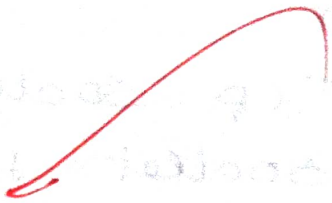
Connected from ('127.0.0.1', 54781)

Recieved: Hello from reverse connector

connector o/p:

Connecting to listener...

Listener replied: Hello from listener



```
client
server = socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
server.bind((HOST,PORT))
server.listen(1)
print("waiting for reverse connection...")
conn, addr = server.accept()
print("connected from:", addr)
message = conn.recv(1024).decode()
print("Received:", message)
conn.send("Hello from listener".encode())
conn.close()
server.close()
```

Connector program: connector.py

```
import socket
HOST = "127.0.0.1"
PORT = 5000
client = socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
print("connecting to listener....")
client.connect((HOST, PORT))
client.send("Hello from reverse connector".
encode())
reply = client.recv(1024).decode()
print("Listener replied:", reply)
client.close()
```

Result:

The TCP socket-based reverse connection demonstration was successfully executed using python.